Lesson 2.6 Guided Notes

Session 1 · Median and Outliers

I can explain why the median is a better measure of center than the mean when a data set has an outlier. · 6.SP.B.5d

Word / Spanish	Meaning — complete the blank	Example
outlier / valor atípico	A value _____ from all the others.	120 in 39, 41, 43, 44, 45, 47, 120
median / mediana	The _____ value of the ordered data.	44
mean / media	The sum of the values _____ by how many values there are.	(2 + 4 + 9) ÷ 3 = 5

Key idea

An outlier pulls the _____ toward it.

The median stays in the _____, so it describes typical data better when there is an outlier.

Worked example

Car counts: 44, 39, 41, 47, 43, 45, 120. What is a typical count?

Order: 39, 41, 43, 44, 45, 47, 120. The outlier is _____.

The median is the 4th value: _____.

Answer: A typical Saturday has about _____ cars.

You try

1. Data: 12, 15, 11, 14, 13, 60, 16.
a. Outlier: _____
b. Median: _____

_____

2. In the data above, 60 changes to 90. What is the new median?

_____

Sum it up: When a data set has an outlier, I use the ___ because ___.

_____

Session 2 · Comparing Distributions

I can compare two distributions using the median and the interquartile range together. · 6.SP.B.5d

Word / Spanish	Meaning — complete the blank	Example
distribution / distribución	How the data values are _____ across their range.	a box plot shows it
center (median) / centro (mediana)	The _____ value of a data set.	median wait 8 min
spread (IQR) / dispersión (rango intercuartil)	How _____ the middle half is.	IQR 7 min

Key idea

Use the _____ to compare typical values and the IQR to compare consistency.

A higher median does not mean _____; only the IQR shows that.

Worked example

Line 1: median 8 min, IQR 7. Line 2: median 13 min, IQR 12. Which line should you join?

Center: Line 1's typical wait is _____ min shorter.

Spread: Line 1's IQR is _____, so its waits are steadier.

Answer: Join Line _____.

You try

1. Group 1: median 80, IQR 4. Group 2: median 84, IQR 22.
a. Higher typical score: _____
b. More consistent: _____

_____

2. Team A: median 30, IQR 8. Team B: median 44, IQR 8. How are they alike? How do they differ?

_____

Sum it up: To compare two distributions, I use the median to ___ and the IQR to ___.

_____

Session 1 · Model, reason, and explain

Median and Outliers · 6.SP.B.5d

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Data A are 4, 5, 6, 7, 8. In data B, replace 8 with 28. Find each mean and median. Which is more resistant to this change, and what does 'typical' mean in your explanation?

A: 4, 5, 6, 7, 8

B: 4, 5, 6, 7, 28

_____

_____

_____

Extend your reasoning

When would the mean of B still be useful?

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Comparing Distributions · 6.SP.B.5d

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Queue A has median 7 minutes and IQR 8 minutes. Queue B has median 10 minutes and IQR 2 minutes. Recommend a queue for a person who values a shorter typical wait, then for a person who values consistent waits.

Queue	Median (min)	IQR (min)
A	7	8
B	10	2

_____

_____

_____

Extend your reasoning

What additional information would help someone who must never wait longer than 15 minutes?

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?